



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df020-2017-android
Operator:	Floris
Analysis Started:	2026-06-18 12:08:38
Analysis Completed:	2026-06-18 12:10:44
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df020\2017\android
Output Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df020\2017\dfdof_output\ft-df020-2017-android
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df020_android.001	controller_android	physical	4.9 GB	2026-06-18 12:09:05
df020_external.001	drone_sd	physical	14.8 GB	2026-06-18 12:09:50

Evidence Source Path

C:\Users\Floris\Documents\ddj_drone\test_mavic_pro\df020\2017\android

Cryptographic Hashes

df020_android.001		controller_android	physical
SHA-256:	b02c8a8b05efbf9fda9bd47a4c87c3c6ab59805df6abdf7058ab164d18ca5f94		
SHA-1:	7568cd5a693349da94f89a106c93dbbfa3ea0bca		

df020_external.001		drone_sd	physical
SHA-256:	46de2036acf691e5c1cbeceb784993e236d61cf7487d6f8cd4057e491a843846		
SHA-1:	d9334ed9dc418533429bb281e20b6e4261067704		

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p1 - controller android]: mmls: no partition table found in df020_android.001; treating as direct partition image
2. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for df020_android.001
3. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for df020_android.001
4. [p2 - controller ios]: source evidence not found
5. [p3 - controller android - drone logs]: no artefacts found
6. [p3 - controller android - images]: no artefacts found
7. [p3 - controller ios]: source evidence not found
8. [p3 - drone flight storage]: source evidence not found
9. [p4 - controller ios]: source evidence not found
10. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
exiftool	9	6	[~] Partial
fls	2	2	[OK]
icat	51	51	[OK]
mmls	2	1	[~] Partial
parser_android	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	6 / 7
Flights With Primary Correlation	0 / 1
Tools Succeeded	4 / 6

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	dji.go.v4	[K] Controller-only	1
Drone Serial Number	08RDE5F001007H	[K] Controller-only	2
Drone Model	Mavic Pro	[K] Controller-only	1
DJI Application Version	4.1.3	[K] Controller-only	1
Account E-Mail	droneforensics@vtoinc.com	[K] Controller-only	1
Product Type	13	[K] Controller-only	1
Last Country Code	US	[K] Controller-only	1
Aircraft Model Code	wm220	[K] Controller-only	1
Controller Serial Number	08TLE4700X0040	[K] Controller-only	1
Last Known Latitude	39.961207508370144	[K] Controller-only	1
Last Known Longitude	-106.21639297141867	[K] Controller-only	1
Account UID	879825329191059456	[K] Controller-only	1
Account Nickname	dronefo...	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: 7a7eca6e632849b01d0d6de04496ad498e2fddadeda8e477b06aab2ebc9de64b
Flight Record Name: DJIFlightRecord_2017-08-29__12-31-05_.csv
Start: 2017-08-29T18:31:05.159000+00:00
End: 2017-08-29T18:38:36.892000+00:00
Duration: 451.7 s

Flight Metrics

Peak Height: 29.7 m
Distance Travelled: 1212.61 m
Video Recording: No
Photo Capture: No
Photos Taken: 0
Recording Duration: 0.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. TripodGPS
5. GoHome
6. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-08-29 18:31:05	ctrl_and:fr (v4.1.3)	Log started	SN: 08RDE5F001007H; App: 4.1.3; Mavic Pro
2017-08-29 18:31:05	ctrl_and:fr (v4.1.3)	Motor turned on	on
2017-08-29 18:37:51	ctrl_and:fr (v4.1.3)	Reached peak height	29.7 m
2017-08-29 18:38:36	ctrl_and:fr (v4.1.3)	Log ended	Dist home: 0.7 m; Height: 1.7 m; Rec: 0.0 s
2017-08-29 18:38:36	ctrl_and:fr (v4.1.3)	Motor turned off	off

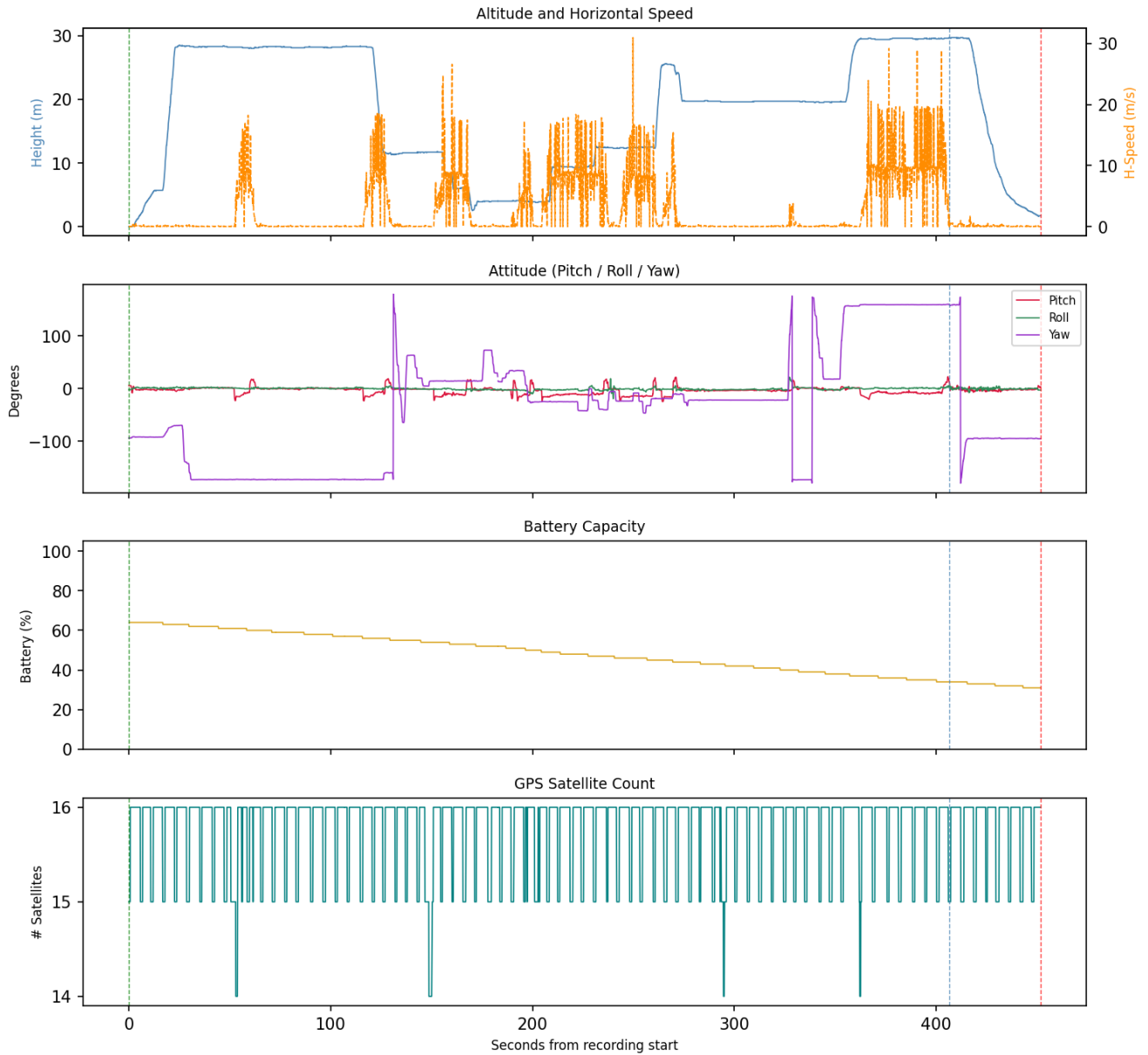
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2164)
p5:6574dfe94ee49b298fa37e736d572d2d2069c0fa013bd4149b94aa5fee16e877	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2164)
p5:68eb24a05eae9a71ef79f0e865dd4b8f928db737655e282b0958b2462c972b02	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2164)
p5:0efec08bdc3058e73f76114f653680f509c8f6f9e8e380ca3057bf7453c8bffd	

Flight Record - Sensor Telemetry

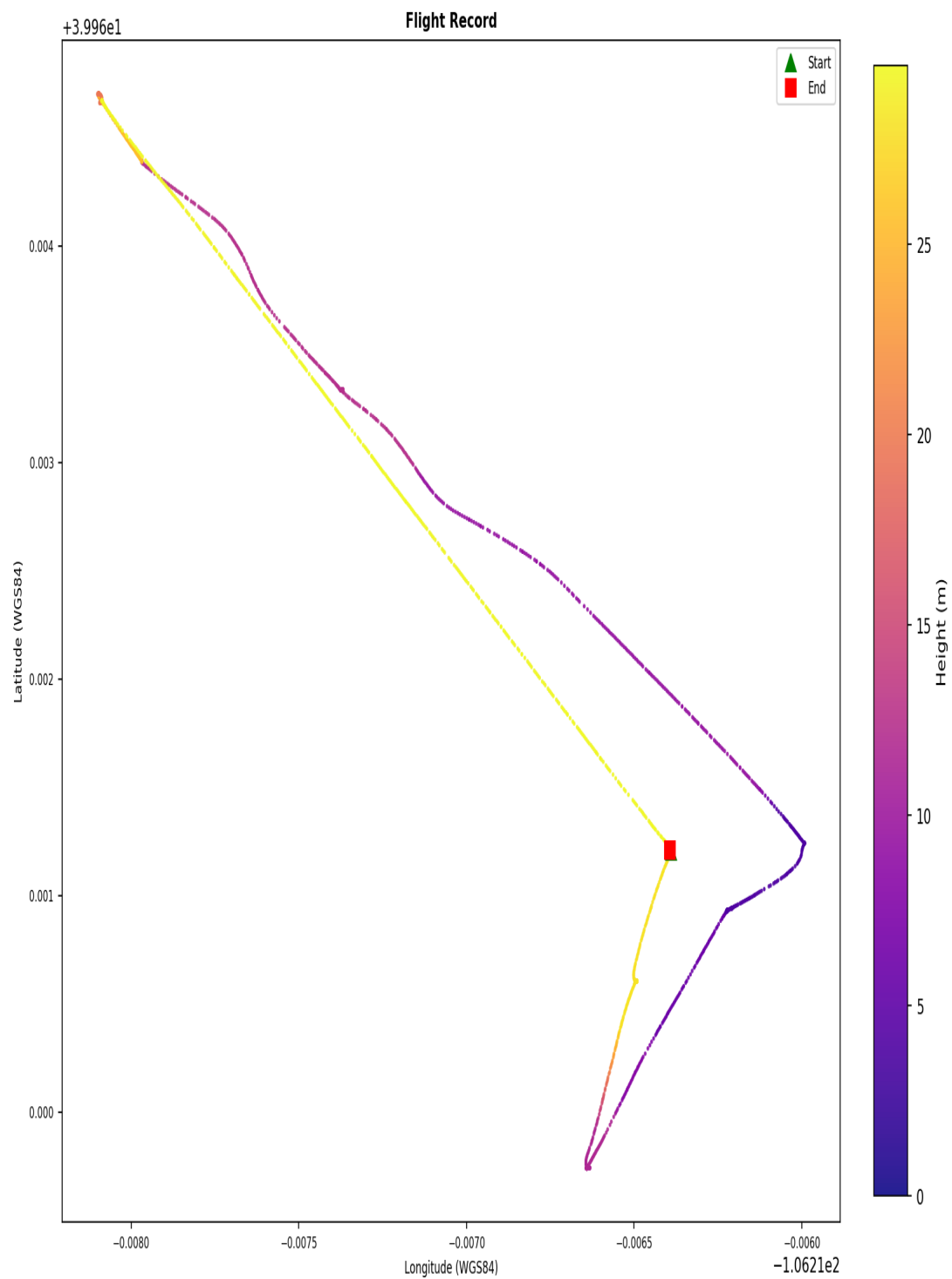
Flight Record Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_01



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	Yes
controller_ios	No
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_android	1	5	0	33	1	0	2
drone_sd	0	0	0	0	0	5	3

Data Quality Notes per Category

Flight Logs (33 files):

- 1 file(s) unreadable (binary)
- formats: 3 bracketed_log, 2 heading_log, 27 unknown

Flight Records (1 files):

- 1 file(s) with row anomalies
- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Images (5 files):

- 2 file(s) contain(s) EXIF GPS coordinates
- 5 file(s) missing (normalisable) EXIF timestamp
- 3 file(s) missing EXIF GPS coordinates

Videos (5 files):

- 4 file(s) contain(s) EXIF GPS coordinates
- 4 file(s) missing (normalisable) EXIF timestamp
- 1 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Flight_logs: 1 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 3 bracketed_log, 2 heading_log, 27 unknown.
- Flight_records: 1 file(s) with row anomalies detected in telemetry data.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

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4. [p2 - controller ios]: source evidence not found
5. [p3 - controller android - drone logs]: no artefacts found
6. [p3 - controller android - images]: no artefacts found
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Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Flight Logs

26-08-2017

- format: heading_log | entries: 1

29-08-2017

- format: heading_log | entries: 4

log-2017-08-25.txt

- format: unknown | entries: 0

log-2017-08-25_1.txt

- format: unknown | entries: 0

log-2017-08-25_10.txt

- format: bracketed_log | entries: 1

log-2017-08-25_11.txt

- format: unknown | entries: 0

log-2017-08-25_2.txt

- format: unknown | entries: 0

log-2017-08-25_3.txt

- format: unknown | entries: 0

log-2017-08-25_4.txt

- format: bracketed_log | entries: 6

log-2017-08-25_5.txt

- format: unknown | entries: 0

log-2017-08-25_6.txt

- format: unknown | entries: 0

log-2017-08-25_7.txt

- format: unknown | entries: 0

log-2017-08-25_8.txt

- format: unknown | entries: 0

log-2017-08-25_9.txt

- format: unknown | entries: 0

log-2017-08-26.txt

- format: unknown | entries: 0

log-2017-08-26_1.txt

- format: unknown | entries: 0

log-2017-08-26_10.txt

- format: unknown | entries: 0

log-2017-08-26_11.txt

- format: unknown | entries: 0

log-2017-08-26_12.txt

- format: unknown | entries: 0

log-2017-08-26_3.txt

- format: unknown | entries: 0

log-2017-08-26_4.txt

- format: bracketed_log | entries: 4

log-2017-08-26_5.txt

- format: unknown | entries: 0

log-2017-08-26_6.txt

- format: unknown | entries: 0

log-2017-08-26_7.txt

- format: unknown | entries: 0

log-2017-08-26_8.txt

- format: unknown | entries: 0

log-2017-08-26_9.txt

- format: unknown | entries: 0

log-2017-08-29.txt

- format: unknown | entries: 0

log-2017-08-29_1.txt

- format: unknown | entries: 0

log-2017-08-29_2.txt

- format: unknown | entries: 0

log-2017-08-29_3.txt

- format: unknown | entries: 0

log-2017-08-29_4.txt

- format: unknown | entries: 0

log-2017-08-29_5.txt

- format: unknown | entries: 0

Flight Records**norm_DJIFlightRecord_2017-08-29__12-31-05_.csv**

- motor_airborne_off: 1 row(s) affected
- contains_no_value: 65 column(s) - APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, CAMERA_INFO.cameraType, CAMERA_INFO.connectState (+60 more)
- contains_constant_value: 145 column(s) - APP_GPS.accuracy, CUSTOM.isVideo, DETAILS.activeTimestamp, DETAILS.aircraftName, DETAILS.aircraftSnBytes (+140 more)

Images**DJI_0001.THM**

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0004.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos**2017_08_29_12_30_57.mp4**

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0004.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0005.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no controller device name found - Device and account information.
2. Note: no controller device found - Device and account information.
3. Note: no firmware version found - Device and account information.
4. Note: no device uuid found - Device and account information.
5. 1 flight record(s) could not be correlated to a drone log (flight_01). Independent corroboration is absent for these flight(s).
6. 5 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

dji.go.v4.xml	Account Data	controller_android	extract_physical
SHA-256:	64c53eedc9aa4e3cc753e12000162ef4b655d582e3db1af8b62805c02076acad		
SHA-1:	d6f66a8f99642c0d13ae81736079f2a281a15eb4		

dji.db	Databases	controller_android	extract_physical
SHA-256:	53a17a371c91168b3b7daaba9cfff6e7df897ef7a508ddb6aec7e89cc037c8144		
SHA-1:	263a550feac460046f2541774cf3977921e5889c		

flysafe_polygon_1860.db	Databases	controller_android	extract_physical
SHA-256:	de615b0a8e225bc1a815b94308e7d1644b648995c43bc0af075f0990adadb90d		
SHA-1:	220f2c8924ad0b196f233dc26b4adfb30367d539		

jpush_local_notification.db	Databases	controller_android	extract_physical
SHA-256:	c785ff31316c80b481d1ba6da796f401253cd17c812b3a41988c8aa9faea89b6		
SHA-1:	3d1cb1b9a2f2d533e0a84d01c7a4e3f9c115c85d		

jpush_statistics.db	Databases	controller_android	extract_physical
SHA-256:	1910cd44fc180faa99bae9c81cfaa1aa2e16163b10f127f36c2e04f3e5896d52		
SHA-1:	31e8c5a5f443bf9b46a26f89d6ce1d62086c128b		

logdb.db	Databases	controller_android	extract_physical
SHA-256:	728e27bd136177543ec18d34832a6a7910b3b73e2b7b0d4e76508c0ff73e934f		
SHA-1:	93396c1b81cad0932535f5d1474fb2d352b3a890		

26-08-2017	Flight Logs	controller_android	extract_physical
SHA-256:	375a055cd6369fb010609dce6d756deb23053e58261facfec09de5f7b5c0aa9e		
SHA-1:	93ecfaf910666a78b10f38714935aec195819d96		

29-08-2017	Flight Logs	controller_android	extract_physical
SHA-256:	3676c792d2406d996b6ede2ac29f96f508160525aabe7f2e662f538497f149		
SHA-1:	a3a42dfd9fbd142d74ad02edeabef9647ed42c1		

log-2017-08-25.txt	Flight Logs	controller_android	extract_physical
SHA-256:	ff6eca14b48406e66a3d5ff258cdef2156e444882d295acbc2252c2860012e23		
SHA-1:	bf4d13821f9dc7dd6bb0deb477d25bdb68019733		

log-2017-08-25_1.txt	Flight Logs	controller_android	extract_physical
SHA-256:	4b5648da477701cba520f542f624820c7dd664b7f85c158aa3ec321784701f66		
SHA-1:	f28ef3dfd5050b7c8c2aded321bbb68748933f14		

log-2017-08-25_10.txt	Flight Logs	controller_android	extract_physical
SHA-256:	6447793ba93be2c9eb6f655bc4e3912673eaac1979496a55675226b9829a296d		
SHA-1:	ab54fc9b83857be07455e6a949f417e4794b2686		

log-2017-08-25_11.txt	Flight Logs	controller_android	extract_physical
SHA-256:	61c8eaa77bf799356bd0502a1e8d2c90cba75f5c9011b07ef53ac0d44a8a29f		
SHA-1:	5b596e6ddf1e222090277d406fd32c46ad3f0f4d		

log-2017-08-25_2.txt		Flight Logs	controller_android	extract_physical
SHA-256:	21a8dc30ff9e13a92edeb8c670986963d28140661c50e1f1f7ff1a0ad074d00a			
SHA-1:	60252a700cda06ae318aed9285955995cd952a7e			

log-2017-08-25_3.txt		Flight Logs	controller_android	extract_physical
SHA-256:	3a8af8e2a9eebca317b4ee10107a6ba59473f820961bee225b4627f72316ef5a			
SHA-1:	43da7a977982078792220af7231b2384abff4316			

log-2017-08-25_4.txt		Flight Logs	controller_android	extract_physical
SHA-256:	6c2e7612f7b1d9b68af8d59edfb772e2e5f77e43474f99d7caba453681306eaf			
SHA-1:	9cf2dd8d09a3fa43ab0696c7a630fd1efbe46790			

log-2017-08-25_5.txt		Flight Logs	controller_android	extract_physical
SHA-256:	db04e0abf4f9028ab99d5ab72e4a4462c0027fcd43da646b3d9c794ecaf03ca8			
SHA-1:	6c3c1f4ef2830e0cbaad079fb9ad42182b78c045			

log-2017-08-25_6.txt		Flight Logs	controller_android	extract_physical
SHA-256:	5c59e73ff66e64a5b981dd0a0ece07edfbb1618d9f04f5e1701f004f46c988b8			
SHA-1:	b4228d0a39ca5c2204bd30d916bdc3256290935c			

log-2017-08-25_7.txt		Flight Logs	controller_android	extract_physical
SHA-256:	61eeef46341ac7123208a92af889d5889b66322b70414763f6386627bc53845f			
SHA-1:	d5a0a61979ba80bd4a2082db10f0c1bf9091e235			

log-2017-08-25_8.txt		Flight Logs	controller_android	extract_physical
SHA-256:	c3449f386001a3d7aad85f27bf7fff6702c54df67e4cea689432242d21f2f991			
SHA-1:	23326d7240a4d87cd2085ef51e9711f469bb18aa			

log-2017-08-25_9.txt		Flight Logs	controller_android	extract_physical
SHA-256:	6e767ad1855dc18d142922966495c5deb5e001dc4e06186ea4bffc050056441			
SHA-1:	04c6cddf879bd091c25915d1cb7ccb7fa32a7273			

log-2017-08-26.txt		Flight Logs	controller_android	extract_physical
SHA-256:	81cdd58b3e3a036127a15e60c39d859aae639b6b0308b1a2c2ae632ce261d5c5			
SHA-1:	7266c203c608f465fbfd6c55be9102c7d9c36004			

log-2017-08-26_1.txt		Flight Logs	controller_android	extract_physical
SHA-256:	57ce8c5b6d63d29f75e6adefea7500e16a24872e22dcd15b49d8ffce4b4eac37			
SHA-1:	e1cef16d4a797d976f9a3d5dccf3cd6ef285b68f			

log-2017-08-26_10.txt		Flight Logs	controller_android	extract_physical
SHA-256:	07b07b7c4aace25e70ded5ed53fabac643f0270e9fce14c2d48dca4c9052cdce			
SHA-1:	98ea5f83cc5daa532b27c19210904555bc1852a1			

log-2017-08-26_11.txt		Flight Logs	controller_android	extract_physical
SHA-256:	e7ae060ee3b40e0c784fb471c6db27ee63229a7c567f2cba589dcb7bb4e57e60			
SHA-1:	2549f936d938debfe400538bf7102b25e882d2bd			

log-2017-08-26_12.txt		Flight Logs	controller_android	extract_physical
SHA-256:	8dd09c962706988b81b295a973af813ac5ca28f1641f117d38c5e3292d8eb845			
SHA-1:	39460cf505756a9320f2903de738f8ce93c8a004			

log-2017-08-26_2.txt	Flight Logs	controller_android	extract_physical
SHA-256:	caade39f7cccc9d0199d02b8fbf5bc653fa214bdbc72a9d36033ffe45746ff3		
SHA-1:	5696f85a5fdc001cf524589b589ab99a80897987		

log-2017-08-26_3.txt	Flight Logs	controller_android	extract_physical
SHA-256:	500ca3ac8b39d7d545270cd8f89063d4af75f568c2c9d8e8170a3f0a17b4e6db		
SHA-1:	a5cfe32919bb06dbfa8becea7f2a5fce7fa5f226		

log-2017-08-26_4.txt	Flight Logs	controller_android	extract_physical
SHA-256:	16c549404c4c210160b8958585d2b0f8c270d36f63143ee52940d326564dad88		
SHA-1:	2342ac8c4819dddef25eb5e1c3540633a1afa058		

log-2017-08-26_5.txt	Flight Logs	controller_android	extract_physical
SHA-256:	33600ac44dad0b8492b9a4bdb487195020dd5a68ff4294ee685b4a5ff8648a68		
SHA-1:	95b4385304e2de92597ffb8bf332d406a8c00abc		

log-2017-08-26_6.txt	Flight Logs	controller_android	extract_physical
SHA-256:	05da66f18db6aabb18a598793262add8e661f9930fc6c391bb623383a8780d91		
SHA-1:	d0cea02bf88928efec6bf94df0ef0b6106be63a0		

log-2017-08-26_7.txt	Flight Logs	controller_android	extract_physical
SHA-256:	2121a1998106f18c6e5934995842c982ba3664d7be7073dd6e7b3d7f961c9a5d		
SHA-1:	666fec7665a04f2642d514967058a06a3c12ce2c		

log-2017-08-26_8.txt	Flight Logs	controller_android	extract_physical
SHA-256:	f0eb3c6995724c102231ba46c9d6e105e70598942f0bf884147423b45e788fe3		
SHA-1:	e29d2d4342ea1ab87173739eb6f014db0c914217		

log-2017-08-26_9.txt	Flight Logs	controller_android	extract_physical
SHA-256:	674c05696699cf4d068ca0d35397301f84d2cea057dab962653de52b3ba0a9b5		
SHA-1:	1c00d4369a55f05d4898fef881a47c675b6ed82b		

log-2017-08-29.txt	Flight Logs	controller_android	extract_physical
SHA-256:	3c81c3ad4b527d2326e8843f43c776ed8b21752cf4ee75ecf1f92cc45d657373		
SHA-1:	6e937f3a3ee343d689aacd2571bcb3837919ab10		

log-2017-08-29_1.txt	Flight Logs	controller_android	extract_physical
SHA-256:	d26743d0737bfceafb8630e67540a72dd9c6a924676450178c5c322bf29cc453		
SHA-1:	4cd02639b42b81a4c8fdc7d93c7a242157ba219a		

log-2017-08-29_2.txt	Flight Logs	controller_android	extract_physical
SHA-256:	9f5e5f6a3a684f12557a686468f799d6037dafb353840d3cef016681bbfc3490		
SHA-1:	7d1ce8a8b2a4e3033e3eae3066f7cd393b4b8346		

log-2017-08-29_3.txt	Flight Logs	controller_android	extract_physical
SHA-256:	b8521e5452d4fd226b23c1e4cb068532fcae97c0f77d4f86fd9d289c2c50fc74		
SHA-1:	b02ee8dbd65ca94ea94116543e5d0ee9a6c010a7		

log-2017-08-29_4.txt	Flight Logs	controller_android	extract_physical
SHA-256:	a03d272f477c0ebb73d15f958a060b1ffa0c3e6ef44da729207d6c813b4ef589		
SHA-1:	e6585568133d1df74c52989905be253346846ade		

log-2017-08-29_5.txt		Flight Logs	controller_android	extract_physical
SHA-256:	8bd579a33bd9246a9452fe243f851292d8b178607d625cd8946082d3aaf229a1			
SHA-1:	88b4283d8696d180cae205760f144935203b5cb9			

DJIFlightRecord_2017-08-29_12-31-05_.txt		Flight Records	controller_android	extract_physical
SHA-256:	aab1ba4daf16dccc2bb3abef2b3d1cf7520a35277d0bcbfd6160561166fab4bf			
SHA-1:	347cc348d3ba1b417cf72b1c0db3c7cc249c8f2d			

DJI_0001.THM		Images	drone_sd	extract_physical
SHA-256:	4e23f027a9017ebbb6a1e2714423dff8816985fc9388b08e48e206e6f4190bd7d			
SHA-1:	c331731ce9baafc7e0d8e98ee1b5eed3017ba821			

DJI_0002.JPG		Images	drone_sd	extract_physical
SHA-256:	d0f284faea570872fd451c0906bd1985e58e152cabe271f7ce80e35fe1b5d142			
SHA-1:	b3753796ce2c07730405516cbb8acd57f2580f13			

DJI_0003.JPG		Images	drone_sd	extract_physical
SHA-256:	da2f39ae8ae220ea0567e60e871da87b1ef7c2f66bf1aefb66d86eac3d1a0dd2			
SHA-1:	c5b47d8f4288258a444aa33e90a8523624be34dc			

DJI_0004.THM		Images	drone_sd	extract_physical
SHA-256:	d6de6e1becc1aac8bd084968d48e43b958a1d91e9c43e7643897cda189800a7f			
SHA-1:	7e21ddd036c409dde759460dc4e2740c7b96da1e			

DJI_0005.THM		Images	drone_sd	extract_physical
SHA-256:	b254d105f743634984b958b82ee4730f71b60783f9511a6585ca9da5e3410704			
SHA-1:	5c49385e227e85aa668aafd0cdfa57b7e51d409f			

2017_08_29_12_30_57.info		Videos	controller_android	extract_physical
SHA-256:	6574dfe94ee49b298fa37e736d572d2d2069c0fa013bd4149b94aa5fee16e877			
SHA-1:	6b033b8489f6787ec0ee8e78ec2e250120dd2711			

2017_08_29_12_30_57.mp4		Videos	controller_android	extract_physical
SHA-256:	7aaac13fa67296028a0b1474afc062c33a353f96b9e624e826138b19bfc404cc			
SHA-1:	2ca6ae7f064ec00e175360762f0da4c19084de28			

DJI_0001.MOV		Videos	drone_sd	extract_physical
SHA-256:	68eb24a05eae9a71ef79f0e865dd4b8f928db737655e282b0958b2462c972b02			
SHA-1:	25040bb7c7a9d18e8afa5683d627971348241a57			

DJI_0004.MOV		Videos	drone_sd	extract_physical
SHA-256:	c28ad8959bcacd1b61c6d5425e89bdfc480d643e315a65ddbd2b8c9422229c67			
SHA-1:	69de46ec683056c48e84ada8104d9db5893f20a9			

DJI_0005.MOV		Videos	drone_sd	extract_physical
SHA-256:	0efec08bdc3058e73f76114f653680f509c8f6f9e8e380ca3057bf7453c8bffd			
SHA-1:	c8cad52303de51cb075924924f82e2538aab6239			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Android

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	1	1	0
Databases	5	0	0
Flight Logs	33	0	0
Flight Records	1	1	1
Videos	2	2	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	5	5	0
Videos	3	3	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-18 12:08:38 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df020_android.001

2026-06-18 12:08:38 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df020_android.001

2026-06-18 12:08:38 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df020_external.001

2026-06-18 12:08:49 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df020_external.001

2026-06-18 12:09:54 parser_android v- rc=n/a

cmd: df020_android.001 C:\Users\Floris\Documents\dfdo_output\ft-df020-2017-android\p2_image_parsing\controller_android_parsed
out: controller_android_parsed

2026-06-18 12:09:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 295
out: packages.list

2026-06-18 12:09:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 66254
out: dji.go.v4.xml

2026-06-18 12:09:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 67103
out: dji.db

2026-06-18 12:09:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 67114
out: jpush_statistics.db

2026-06-18 12:09:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 67112
out: flysafe_polygon_1860.db

2026-06-18 12:09:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 67135
out: jpush_local_notification.db

2026-06-18 12:09:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 66760
out: logdb.db

2026-06-18 12:09:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 1296
out: log-2017-08-25.txt

2026-06-18 12:09:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 927
out: log-2017-08-26.txt

2026-06-18 12:09:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 926
out: log-2017-08-29.txt

2026-06-18 12:09:55 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 1300

out: log-2017-08-25_1.txt

2026-06-18 12:09:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 929

out: log-2017-08-26_1.txt

2026-06-18 12:09:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2870

out: log-2017-08-29_1.txt

2026-06-18 12:09:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 1303

out: log-2017-08-25_2.txt

2026-06-18 12:09:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 930

out: log-2017-08-26_2.txt

2026-06-18 12:09:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2871

out: log-2017-08-29_2.txt

2026-06-18 12:09:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 1311

out: log-2017-08-25_3.txt

2026-06-18 12:09:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 933

out: log-2017-08-26_3.txt

2026-06-18 12:09:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2874

out: log-2017-08-29_3.txt

2026-06-18 12:09:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 1313

out: log-2017-08-25_4.txt

2026-06-18 12:09:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 934

out: log-2017-08-26_4.txt

2026-06-18 12:09:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2812

out: log-2017-08-29_4.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 1315

out: log-2017-08-25_5.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 932

out: log-2017-08-26_5.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2841

out: log-2017-08-29_5.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2309

out: log-2017-08-25_6.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 945

out: log-2017-08-26_6.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2322

out: log-2017-08-25_7.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2814

out: log-2017-08-26_7.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2334

out: log-2017-08-25_8.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 897

out: log-2017-08-26_8.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2342

out: log-2017-08-25_9.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2817

out: log-2017-08-26_9.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2365

out: log-2017-08-25_10.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2363

out: log-2017-08-26_10.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 909

out: log-2017-08-25_11.txt

2026-06-18 12:09:57 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 931

out: log-2017-08-26_11.txt

2026-06-18 12:09:58 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2823

out: log-2017-08-26_12.txt

2026-06-18 12:09:58 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 921

out: 26-08-2017

2026-06-18 12:09:58 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2824

out: 29-08-2017

2026-06-18 12:09:58 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2828

out: DJIFlightRecord_2017-08-29__12-31-05_.txt

2026-06-18 12:09:59 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2825

out: 2017_08_29_12_30_57.mp4

2026-06-18 12:10:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df020_android.001 2826

out: 2017_08_29_12_30_57.info

2026-06-18 12:10:07 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 8197

out: DJI_0001.MOV

2026-06-18 12:10:14 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 8198

out: DJI_0002.JPG

2026-06-18 12:10:14 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 8199

out: DJI_0003.JPG

2026-06-18 12:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 8200

out: DJI_0004.MOV

2026-06-18 12:10:27 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 8201

out: DJI_0005.MOV

2026-06-18 12:10:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 9222

out: DJI_0001.THM

2026-06-18 12:10:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 9223

out: DJI_0004.THM

2026-06-18 12:10:38 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 9224

out: DJI_0005.THM

2026-06-18 12:10:39 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-08-29__12-31-05_.txt DJIFlightRecord_2017-08-29__12-31-05_.csv

out: DJIFlightRecord_2017-08-29__12-31-05_.csv

2026-06-18 12:10:40 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_12_30_57.mp4

out: 2017_08_29_12_30_57_exif.json

2026-06-18 12:10:40 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM

out: DJI_0001_exif.json

2026-06-18 12:10:40 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.JPG

out: DJI_0002_exif.json

2026-06-18 12:10:40 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.JPG

out: DJI_0003_exif.json

2026-06-18 12:10:41 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.THM

out: DJI_0004_exif.json

2026-06-18 12:10:41 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.THM

out: DJI_0005_exif.json

2026-06-18 12:10:42 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MOV

out: DJI_0001_exif.json

2026-06-18 12:10:42 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MOV

out: DJI_0004_exif.json

2026-06-18 12:10:42 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MOV

out: DJI_0005_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.